Incident Lifecycle Automation in ServiceNow

Project Design Phase-II • Technology Stack (Architecture & Stack)

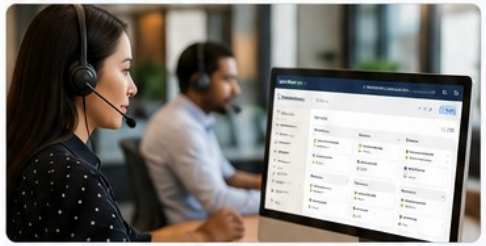
Phase 1: Requirement Analysis & Planning

Epic • Complexity: Medium • Duration: 1h 0m

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Overview



In many organizations, IT incidents are reported through emails, phone calls, or informal communication channels. This leads to poor tracking, lack of proper classification, delayed escalations, and minimal visibility into resolution progress. Without structured **Incident Management**, issues may remain unresolved, SLAs may be breached, and root cause tracking becomes difficult.

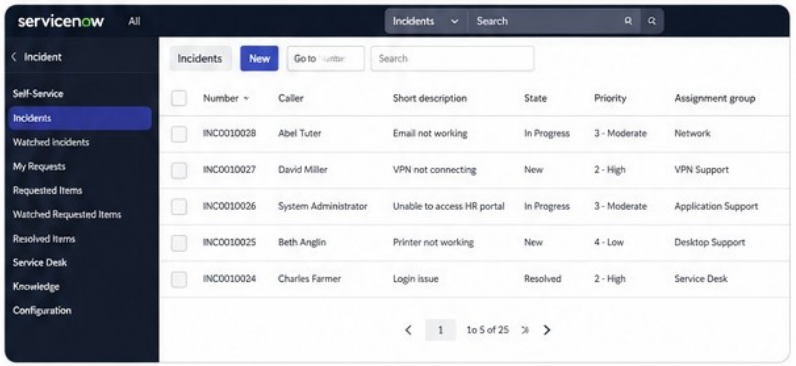
The objective of this project is to automate and standardize the Incident Management iifecycle using ServiceNow's Incident module.

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Description

This phase involves understanding the current incident handling process, identifying pain points, defining requirements, and planning the automation approach in ServiceNow.

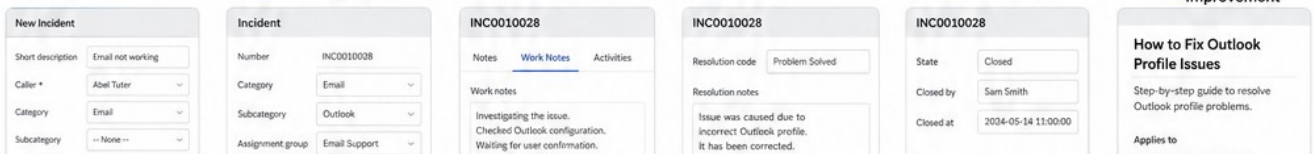


Key Goals

- Enable Service Desk agents to create and classify incidents.
- Integrate Knowledge Management for faster resolution.
- Allow reassignment to appropriate support groups.
- Enable Level 2 technicians to investigate and resolve incidents.
- Create emergency change requests directly from incidents.
- Allow child incident creation for tracking recurring issues.
- Generate knowledge articles from resolved incidents.
- Improve SLA compliance and service visibility.

Incident Lifecycle Flow in ServiceNow

1 Incident Creation → 2 Classification & Assignment → 3 Investigation → 4 Resolution → 5 Follow-up & Closure → 6 Knowledge & Continuous Improvement



1. Overview

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Objective: Automate and standardize the Incident Management lifecycle using ServiceNow's Incident module.

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2. Incident Lifecycle Flow

Lifecycle Stage	Automation / Outcome
1. Incident Creation	Incident is reported by a user or Service Desk agent and captured with key details.
2. Classification & Assignment	Incident is categorized, prioritized, and assigned to the appropriate support group.
3. Investigation	Level 2 technicians investigate using diagnostics, work notes, and knowledge articles.
4. Resolution	The issue is resolved and resolution notes are recorded.
5. Follow-up & Closure	SLA compliance is verified and the incident is closed.
6. Knowledge & Continuous Improvement	Resolved incidents can produce knowledge articles for future reference.

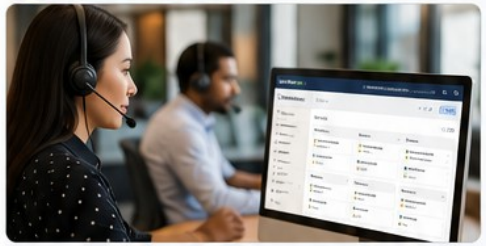
3. Technical Architecture

The architecture below is presented as a ServiceNow-centered cloud solution. It maps the supplied template requirements to the Incident Management lifecycle, including user interaction, application logic, knowledge, change and child incidents, data persistence, integrations, and SLA visibility.

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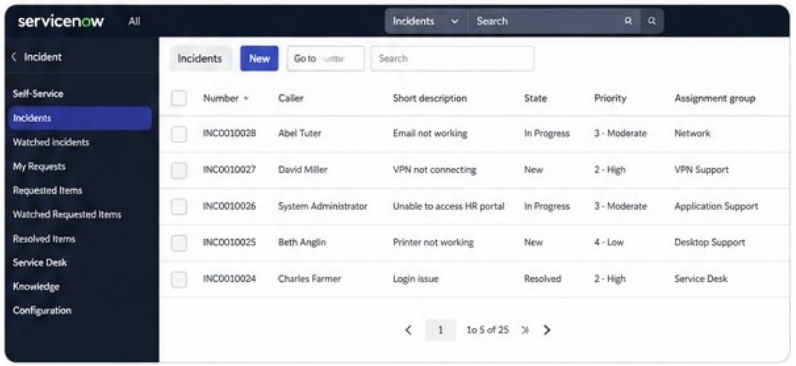
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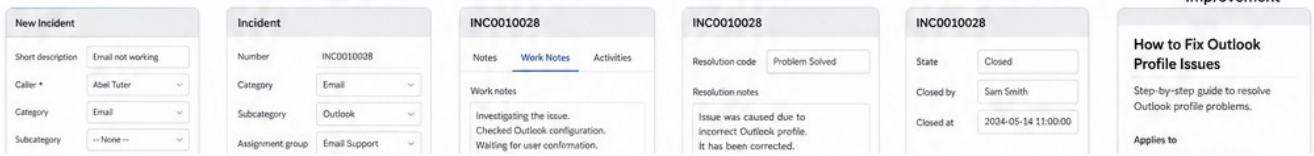


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Components & Technologies

S.No	Component	Description	Technology / Platform
1	User Interface	Service Desk and technician interface for incident creation, classification, investigation and closure.	ServiceNow web UI / Workspace
2	Application Logic-1	Incident creation, classification, prioritization, assignment, investigation, resolution and closure workflows.	ServiceNow Incident Management / Flow Designer
3	Application Logic-2	Knowledge lookup and knowledge-article generation from resolved incidents.	ServiceNow Knowledge Management
4	Application Logic-3	Emergency change and child-incident actions initiated from an incident.	ServiceNow Change / Incident capabilities
5	Database	Incident records, work notes, assignments, SLA data and resolution details.	ServiceNow platform data store
6	Cloud Database	Cloud-hosted persistence provided by the ServiceNow platform.	ServiceNow cloud platform
7	File Storage	Attachments and supporting incident evidence, where required.	ServiceNow attachment storage
8	External API-1	Optional interface to enterprise notification or integration services.	IntegrationHub / REST APIs (if applicable)
9	External API-2	Optional interface to other enterprise systems for incident context or automation.	REST / SOAP integrations (if applicable)
10	Machine Learning Model	Not mandatory in the supplied requirements; may be added for classification or recommendations.	Not applicable in current scope
11	Infrastructure	Application deployment and hosting.	ServiceNow cloud platform

4. Application Characteristics

Characteristic	Description	Technology / Approach
Open-Source Frameworks	The supplied requirements do not mandate an open-source application framework.	Not specified
Security	Role-based access and controlled permissions should restrict incident creation, assignment, updates and administrative actions.	ServiceNow roles / ACLs / IAM controls
Scalable Architecture	A cloud platform supports standardized workflows across Service Desk and Level 2 support groups.	ServiceNow cloud architecture
Availability	Centralized cloud service provides shared access to incidents, workflows, knowledge and reporting.	ServiceNow cloud platform
Performance	Structured forms, workflow automation, knowledge search and dashboards reduce manual handling and improve visibility.	ServiceNow platform capabilities

5. Automation Use Cases

Knowledge Management: Agents can search existing knowledge during incident handling, while resolved incidents can be converted into reusable knowledge content.

Reassignment & Escalation: Incidents can move to the appropriate support group and Level 2 technicians for investigation.

Emergency Change: An emergency change request can be created directly from an incident when restoration requires a change.

Child Incidents: Child incidents can be created to track recurring or related issues while retaining the parent incident relationship.

SLA Visibility: Incident and SLA information can be surfaced in dashboards and reports to support compliance monitoring.

6. Expected Outcomes

- Standardized incident intake and classification.
- Faster resolution through knowledge reuse and structured investigation.
- Clear ownership and reassignment across support groups.
- Controlled emergency change and child-incident tracking.
- Better SLA compliance monitoring and service visibility.
- A continuous-improvement loop through knowledge articles generated from resolved incidents.

7. References

- C4 Model: https://c4model.com/
- IBM Cloud Architecture: https://www.ibm.com/cloud/architecture
- AWS Architecture: https://aws.amazon.com/architecture